

2 Introduction to SQL Queries

1. What is SQL?

SQL (Structured Query Language) is the language used to communicate with a database.

Just like:

- Humans speak English.
- Computers understand programming languages.
- Databases understand SQL.

We use SQL to ask questions to the database.

Example questions:

- What are all the employees?
 - What is today's sales?
 - How many customers do we have?
 - Which machine has maximum downtime?
-

2. Where is the Data?

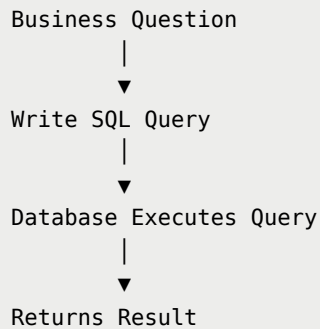
```
Database
|
├─ Table 1
├─ Table 2
└─ Table 3
```

Example

```
Database
|
├─ Employees
├─ Orders
└─ Customers
```

The actual information is stored inside **tables**.

3. How SQL Works



Example

Question:

| How many employees are there?

SQL

```
SELECT COUNT(*)  
FROM Employees;
```

Database

↓

Result

250

4. What is an SQL Query?

An SQL Query is simply a **question** written in SQL language.

Example

Question

Show all employees.

SQL

```
SELECT *  
FROM Employees;
```

The database answers by returning the data.

5. SQL Query Does NOT Change Data

The instructor mentions something very important.

A SELECT query

✓ Reads data

✗ Does NOT

- Delete rows
- Update rows
- Insert rows
- Change table structure

It is completely safe.

Example

```
SELECT *  
FROM Employees;
```

Nothing changes in the database.

6. SQL Query Clauses

A SQL query is made up of different parts called **Clauses**.

Think of them as building blocks.

```
SELECT  
FROM  
WHERE
```

GROUP BY
HAVING
ORDER BY
JOIN
DISTINCT
TOP

Each clause has a different job.

7. SQL Clauses and Their Purpose

Clause	Purpose
SELECT	Choose columns
FROM	Choose table
WHERE	Filter rows
JOIN	Combine tables
GROUP BY	Group similar data
HAVING	Filter groups
ORDER BY	Sort data
DISTINCT	Remove duplicates
TOP	Return first N rows

8. Simplest SQL Query

Every beginner starts with

```
SELECT *  
FROM Employees;
```

Meaning

```
SELECT  
↓  
Choose columns  
  
*  
↓  
All columns
```

```
FROM
↓
Which table?

Employees
```

Result

```
ID
Name
Salary
Department
```

9. Flow of Asking the Database

```
You
|
| Ask Question
▼
SELECT ...
|
▼
Database
| Searches Data
▼
Returns Result
|
▼
You read the answer
```

10. Important Point

A database **never** understands English.

It understands SQL.

Instead of saying

| Show me all employees

We write

```
SELECT *  
FROM Employees;
```

11. Real World Example

Imagine a school.

Question

How many students are in Class 10?

You don't ask every student.

You ask the school database.

```
SELECT COUNT(*)  
FROM Students  
WHERE Class = 10;
```

The database searches thousands of records and returns the answer in milliseconds.

12. SQL Learning Roadmap (As per Instructor)

The instructor says we'll learn these one by one.

SQL Query

```
|  
|— SELECT  
|— FROM  
|— DISTINCT  
|— TOP  
|— WHERE  
|— JOIN
```

- └ GROUP BY
- └ HAVING
- └ ORDER BY

Don't try to memorize everything today.

Today's focus is only understanding what a query is.

Interview Questions

Q1. What is SQL?

SQL is a language used to communicate with relational databases and retrieve or manipulate data.

Q2. What is an SQL Query?

An SQL query is a statement written in SQL to request information from a database.

Q3. What does SELECT do?

SELECT retrieves data from one or more tables.

Q4. Does SELECT modify data?

No. SELECT only reads data.

Q5. Where is data stored?

Inside tables, which are stored inside a database.

Practice

Suppose you have this table.

Employees		
ID	Name	Salary
1	John	5000

2	Alice	7000
3	David	6000

Practice these queries:

```
SELECT *  
FROM Employees;
```

```
SELECT Name  
FROM Employees;
```

```
SELECT Salary  
FROM Employees;
```

```
SELECT ID, Name  
FROM Employees;
```

Don't worry about `WHERE` or `ORDER BY` yet—you'll learn those in the next lessons.

One-Page Revision Summary

SQL = Language for communicating with a database.

Database

↓

Tables

↓

Data

Question

↓

Write SQL Query

↓

Database Executes

↓

Returns Result

SELECT = Read data only

SQL Query = Collection of clauses

Main clauses:

SELECT	→ Choose columns
FROM	→ Choose table
WHERE	→ Filter rows
JOIN	→ Combine tables
GROUP BY	→ Group data
HAVING	→ Filter groups
ORDER BY	→ Sort data
DISTINCT	→ Remove duplicates
TOP	→ First N rows
